Architecture (ex. ARC)

Exercises Solutions Architecture des ordinateurs

1 Architecture

1.1 Stack-Architecture

a) -
b) 7 explicit fetch and none implicit
c) 7 explicit fetch, 4 implicit fetch, 4 implicit store

1.2 Stack-Architecture

a) -
b) 7 explicit fetch
c) 7 explicit fetch (with compiler optimizations)
7 explicit fetch, 1 implicit store, 1 implicit fetch (without compiler optimizations)
d) 7 explicit fetch, 3 implicit store, 3 implicit fetch (without compiler optimizations)

2 Single-Cycle RISC-V

2.1 Fonctionnement du processeur à cycle unique

- `PCScr = '0'`
- `RegWrite = '1'`
- `ImmScr[1:0] = "xx"`
- `ALUSrc = '0'`
- `ALUControl[2:0] = "010"`
- `MemWrite = '0'`
- `ResultScr = '0'`

2.2 Prolonger le mono-cycle avec l'instruction `jal`

- $PCScr =' 0'$
- $RegWrite =' 1'$
- $ImmScr[1:0] = "xx"$
- $ALUSrc =' 0'$
- $ALUControl[2:0] = "010"$
- $MemWrite =' 0'$
- $ResultScr =' 0'$

2.3 Performance du processeur à mono-cycle

$$T_{program_single_cycle} = 75sec$$